

The background of the slide features a dark red color with a faint, large-scale watermark of the Chicago Booth crest. The crest includes a shield with an eagle and a banner above it that reads "Crescat Scientia Excolatur".

COMPETITIVE PRICING

Overview

2

Pricing is a balancing act: **value**, **costs** & **competition**

└──┬──┘ └──┬──┘
day 1 day 2

Overview

2

Pricing is a balancing act: **value**, **costs** & **competition**

└──┬──┘ └──┬──┘
day 1 day 2

- Firms under-estimate value of competitor information (see [Kim 2022](#))
 - ▣ Many unaware of competitors' prices
 - ▣ But, 20% adjust their prices when competitor info provided!
- There are exceptions
 - ▣ Flynn Restaurant Group gets monthly reports on competitor pricing

See [“How America's Largest Restaurant Franchisee Decides When to Raise Prices,”](#) by Heather Haddon, WSJ, May 21, 2023

- Let's explore how competitor decisions can impact a market ...

Overview

The Brokerage Price War

3

Robinhood app initiated “zero commissions” in 2013 to profit on orderflow and interest on cash...



Robinhood	E*Trade	Schwab	Scottrade
\$0.00 Per Trade Commission	\$7.99-\$9.99 Per Trade Commission	\$8.95 Per Trade Commission	\$7.00 Per Trade Commission
\$0.00 Minimum Deposit	\$500 Minimum Deposit	\$1,000 Minimum Deposit	\$500 Minimum Deposit

Overview

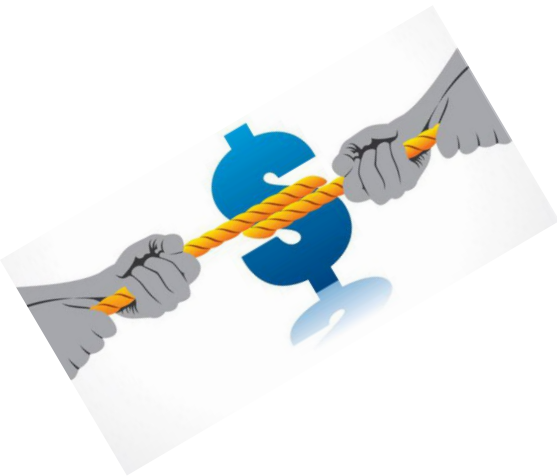
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Charles Schwab, Fidelity Escalate Brokerage Price War

Money managers make **hundreds more ETFs free** to trade on their platforms

Wall Street Journal, February 12, 2019



Schwab Strikes Blow in Brokerage Price War

...**eliminate commissions** on online stock trades, one of most dramatic moves yet in a broad-based price war that is crimping profitability across the financial sector.

Wall Street Journal, October 2, 2019

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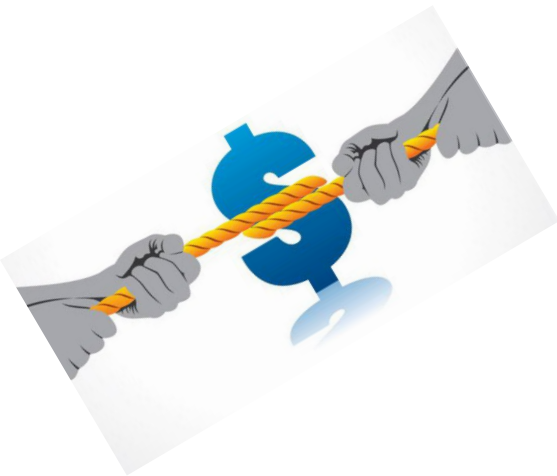
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Incumbents reacted by lowering their prices!

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Incumbents reacted by lowering their prices!

see also EV price war in China: [“Bloodbath looms in China's overcrowded EV industry.”](#)
Straits times, 05-07-2024.

Overview

Pricing A.I. at German gas stations

4

Price-Point

Artificial-intelligence software is being used by some gas stations to constantly adjust prices to maximize sales or margins. Here's how it works:



Step 1

Build a database of historical transactions to teach the software about market dynamics; add competitors' info.



Step 2

Connect software to live feeds of purchase data and other variables such as weather and traffic.



Step 3

Software compares live data to historical numbers to predict demand linked to prices.



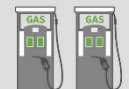
Step 4

Owner sets strategy for each fuel at each station, including preferred balance between volume and margin and constraints such as minimum price.



Step 5

Algorithms determine price for each fuel and automatically adjust pumps throughout the day.



Step 6

Transactions in reaction to those prices fed back into system to generate new predictions and prices.

“When the Machines Learn to Price-Gouge”

Mike Bird, WSJ, Sept. 29, 2020



- Wide adoption of pricing AI by German gas station in 2017

Overview

Pricing A.I. at German gas stations

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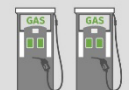
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- ❑ Wide adoption of pricing AI by German gas station in 2017
- ❑ In duopoly markets:
 - ❑ If only one adopted, no impact
 - ❑ If both adopted → large increases in price
→ 30% increase in margins

See: [Assad, Clark, Ershov and Xu \(2023\)](#)

Overview

Pricing A.I. at German gas stations

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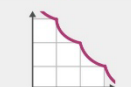
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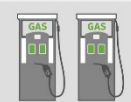
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Automated competitive reactions led to higher prices!

- Wide adoption of pricing AI by German gas station in 2017
- In duopoly markets:
 - ▣ If only one adopted, no impact
 - ▣ If both adopted → large increases in price
→ 30% increase in margins

See: [Assad, Clark, Ershov and Xu \(2023\)](#)

Overview

5

So today:

- Anticipating **competitive response**

- Using **EVC**

- Thinking Like Your Competitor

First half lecture; use EVC framework to think like our competitor. Why? want to understand when we take a radical move with our price, how does it change the competitor's reaction in their pricing.

Pricing optimally requires us to step out of our thinking, and put ourselves in customers and competitors shoes.

Second half; read the Beauregard Textile. Play a pricing game. What happens in equilibrium.

- Modeling strategic price interactions between competitors

- Beauregard Textiles

- Strategic price interactions

- Reaction Functions (**Game Theory**)

- Potentially perplexing **equilibrium** outcomes

Competitive Price Changes and EVC

6

For some customer i

$$EVC_i \equiv \text{Maximum } WTP_i = DV_i + R_i$$

R_i is closest competitor's price

Competitive Price Changes and EVC

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- As competitor increases (decreases) her price:

Competitive Price Changes and EVC

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For some customer i

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- As competitor increases (decreases) her price:
 - EVC of our product increases (decreases)
 - In turn, our demand curve shifts out (in)
- which changes our pricing incentives.

Competitive Price Changes and EVC

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□ As competitor increases (decreases) her price:

➤ EVC of our product increases (decreases)

➤ In turn, our demand curve shifts out (in)

which changes our pricing incentives.

Our demand shift out,
prices (WTP increase)

Now let's put ourselves into our competitors' shoes...

7 EVC to Assess Price Reactions

Hypothetical case study of new product launch at Olympus



Olympus Objectives

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- Olympus planning new product launch:
Olympus, 12.2MP, DSLR camera

← see “competitor
simulation” handout
on Canvas

- Objectives
 - 1 Assess the incumbent market
 - 2 Assess Olympus’ launch price incentives

Market Analysis: Pre Olympus Launch

1 Assess Incumbent Market

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Currently, there are 2 products already in the market:

1. Canon, 5MP, non-DSLR: \$280
2. Sony, 5MP, non-DSLR: \$250

Can we help Olympus derive demand for incumbent brands?

Market Analysis: Pre Olympus Launch

1 Assess Incumbent Market

10

Base model: {Canon, 5 MP, no DSLR}

	Consumer Preferences		
Features	Segment A	Segment B	Segment C
Base Model	\$ 350	\$ 280	\$ 320
MP: 10	\$ 100	\$ 28	\$ 33
MP: 12.2	\$ 120	\$ 40	\$ 60
Sony	\$ (50)	\$ 40	\$ (67)
Olympus	\$ 100	\$ 48	\$ 33
DSLR	\$ 50	\$ 28	\$ 27
Segment size	40%	20%	40%

These
are
utilities
(in \$)

Conjoint Analysis Results
(marketing research on customer)

Market Analysis: Pre Olympus Launch

1 Assess Incumbent Market

11

	Segment A	Segment B	Segment C
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Market Analysis: Pre Olympus Launch

1 Assess Incumbent Market

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□ Incumbent products already in the market are:

- Canon base model at \$280
- Sony base model at \$250

Only thing that
differentiates Canon and
Sony is BRAND. that's all.

□ What is EVC for Canon pre Olympus launch?

Canon Demand

Sony at \$250, Olympus **not** offered

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Sony Base Model

Price = \$250

What is demand for Canon Base Model?

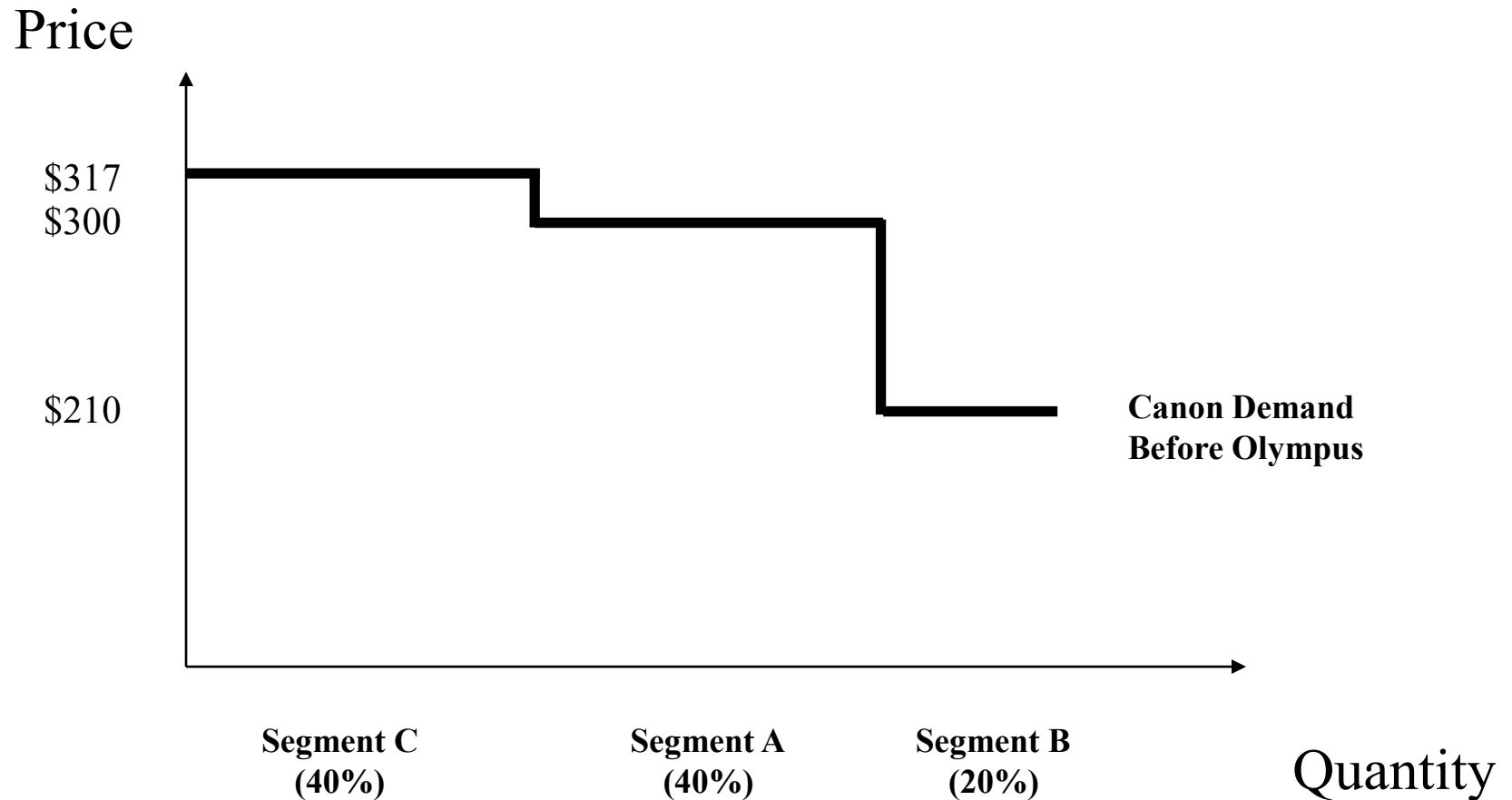
	Segment A	Segment B	Segment C
Ref Brand	Sony	Sony	Sony
Ref Value	\$250	\$250	\$250
Diff Value			
Meg	-	-	-
Brand	\$50	\$(40)	\$67
DSLR	-	-	-
EVC	\$300	\$210	\$317

Segment A and C prefers Canon compared to Segment B, hence \$ positive value for brand.

Canon Demand

EVC demand **before** Olympus launches

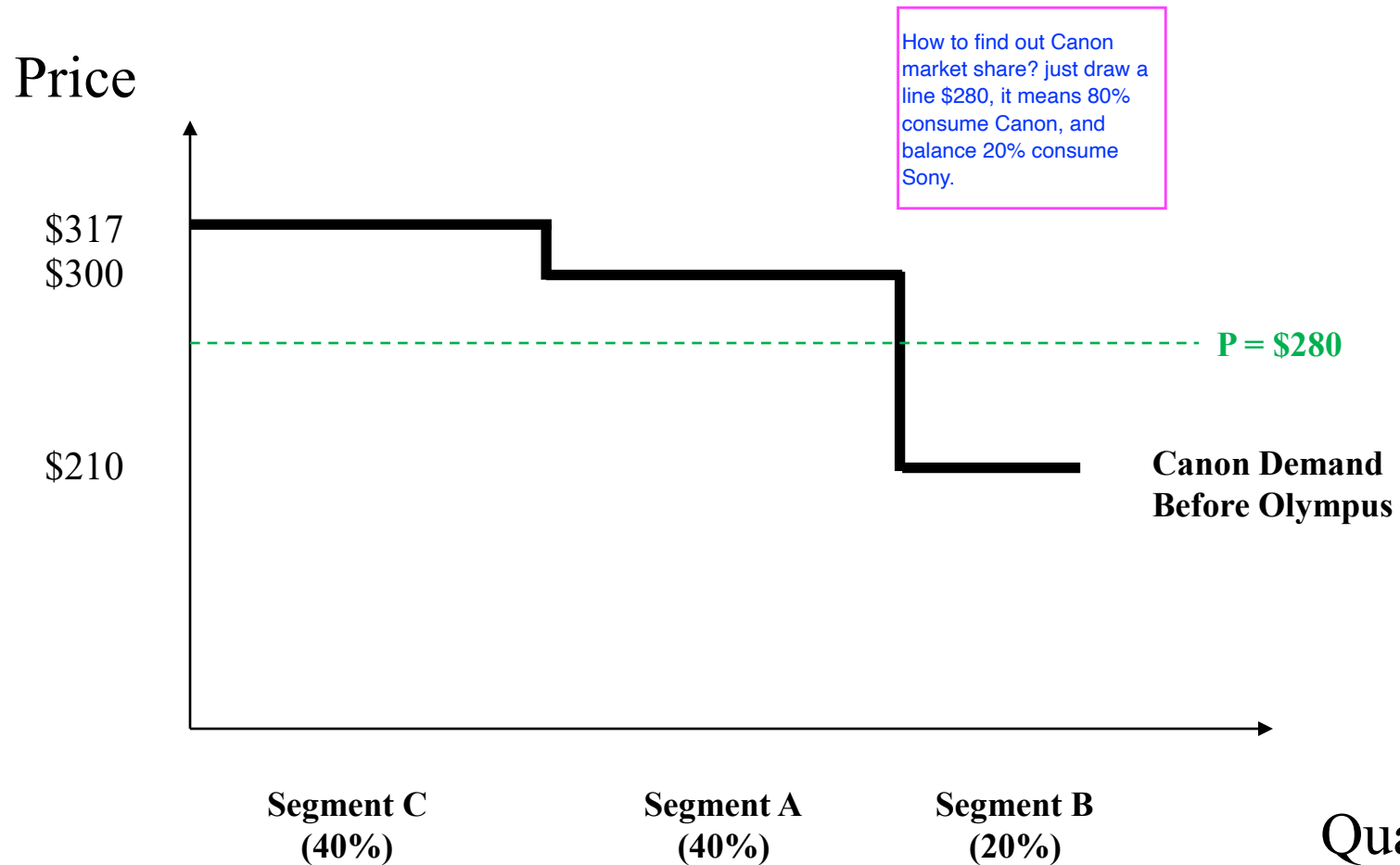
13



Canon Demand

EVC demand **before** Olympus launches

13



Market Analysis: Olympus Launch

2 Assess Olympus launch price incentives

14

- Determine demand for Olympus 12.2MP DSLR

(new firm, new product)

- Assess pricing strategies for Olympus

This is a **Market Simulation**:

- Impact on Canon and Sony
- Likely reactions from Canon and Sony

Market Analysis: Olympus Launch

2 Assess Olympus launch price incentives

14

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This is a **Market Simulation**:

- Impact on Canon and Sony
- Likely reactions from Canon and Sony

What's Olympus' reference product?

Demand for Olympus 12.2 MP DSLR

15

- First, determine Olympus' reference product in each segment
- Can just use the demand curves
- OR ...
 - ▣ If Olympus not offered, will segment pick base Sony or Canon?
 - Canon-Sony Price: \$30
 - If the Diff. Value $>$ \$30 then buy Canon
 - ▣ Diff. Value of Canon vs. Sony
 - Seg. A: $+\$50 > \30 so buy Canon
 - Seg. B: $-\$40 < \30 so buy Sony
 - Seg. C: $+\$67 > \30 so buy Canon
- This gives us the Reference Brand for each segment.

Demand for Olympus 12.2 MP DSLR

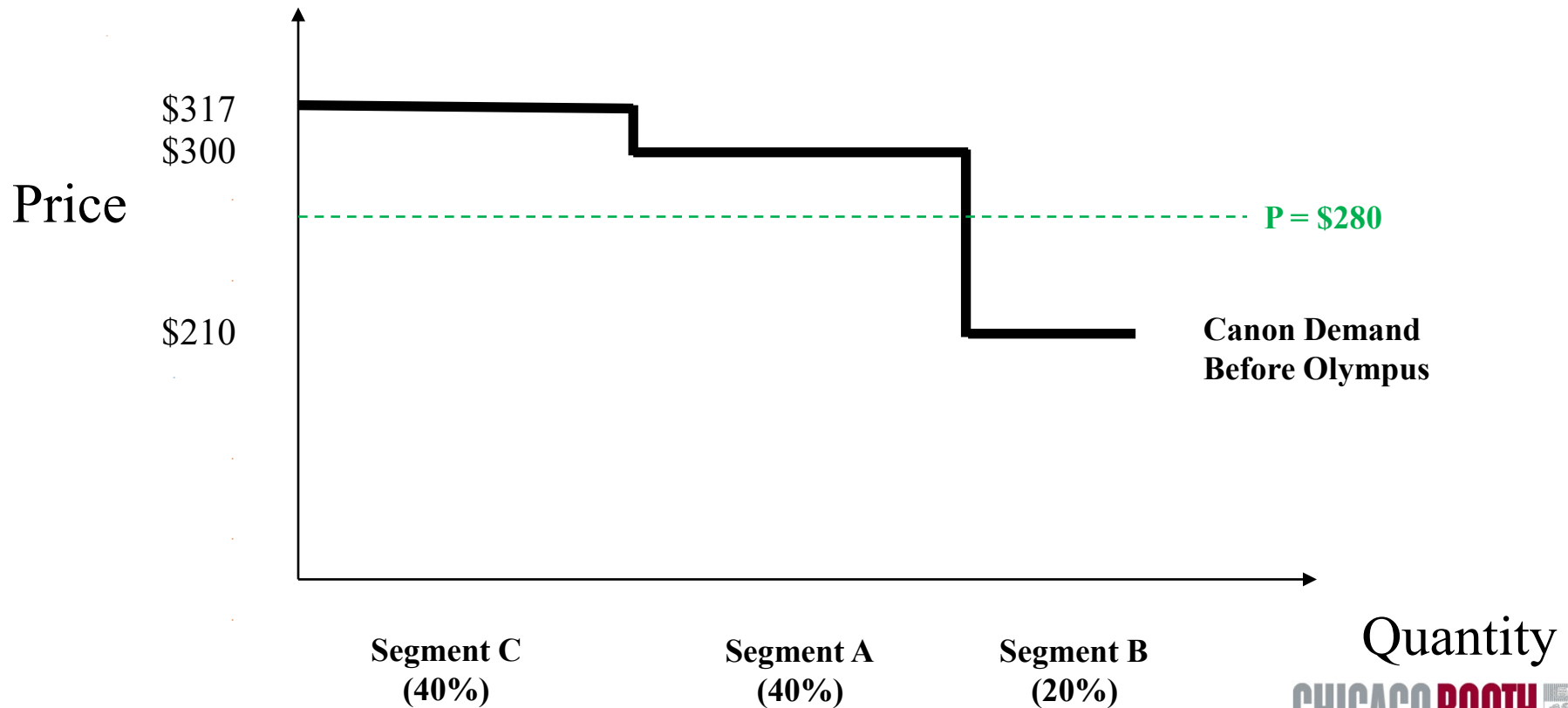
15

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Demand for Olympus 12.2 MP DSLR

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Demand for Olympus 12.2 MP DSLR

Canon Base \$280, Sony Base \$250

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Canon Base Model	Sony Base Model
Price = \$280	Price = \$250
Segments A & C prefer Canon	
Segment B prefers Sony	
What is demand for Olympus 12.2 MP, DSLR Model?	

	Segment A	Segment B	Segment C
Ref Brand	Canon	Sony	Canon
Ref Value	\$280	\$250	\$280

Diff Value			
Meg	\$120	\$ 40	\$ 60
Brand	\$100	\$ 8	\$ 33
DSLR	\$ 50	\$ 28	\$ 27

How much added value if they chose Olympus over their own reference product.

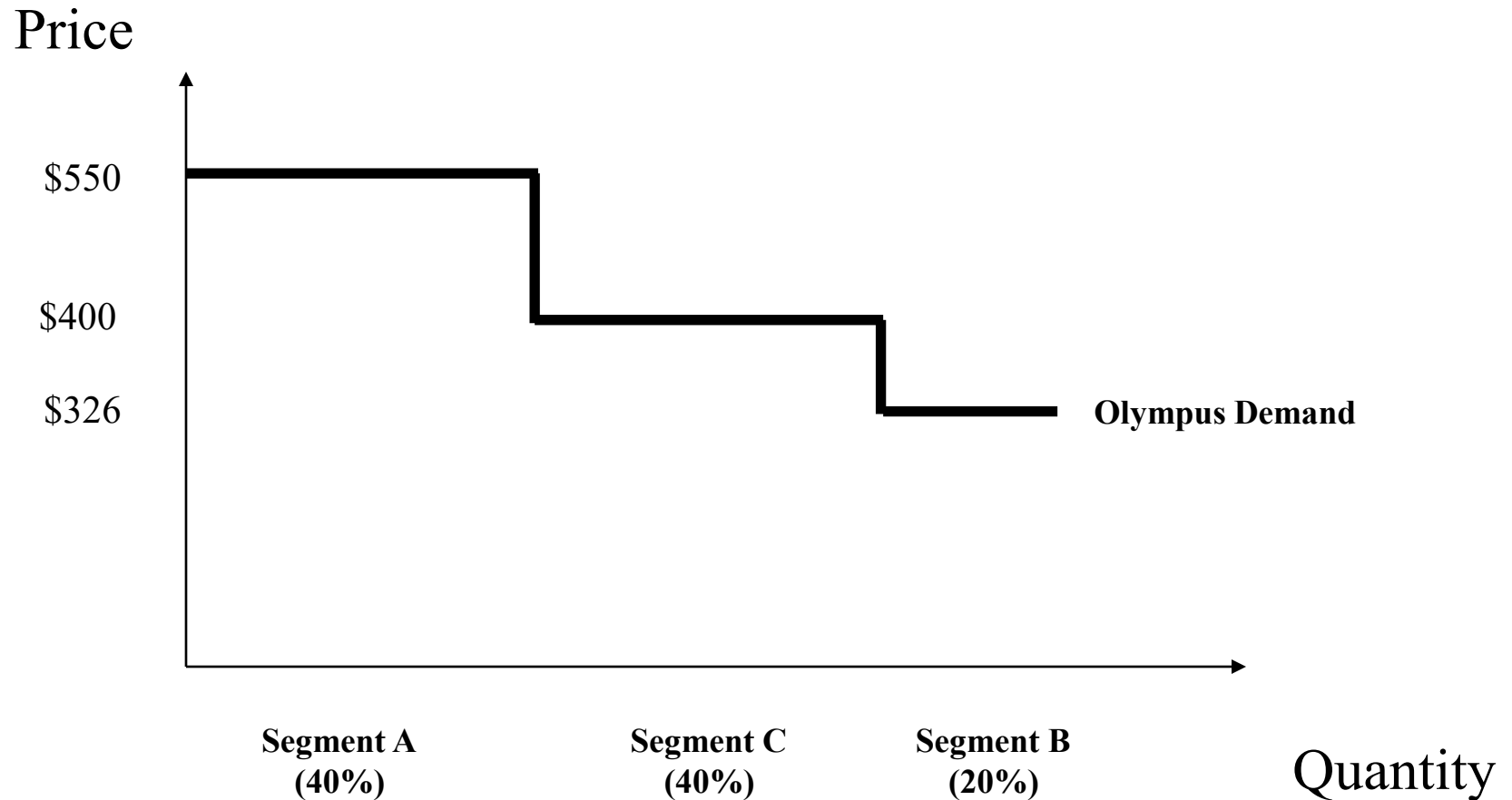
we can tell Olympus that even though there is a reference \$280 canon in the market, people are willing to pay \$326-550 for your product.

EVC	\$550	\$326	\$400
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Olympus Demand

EVC demand when it launches against incumbents

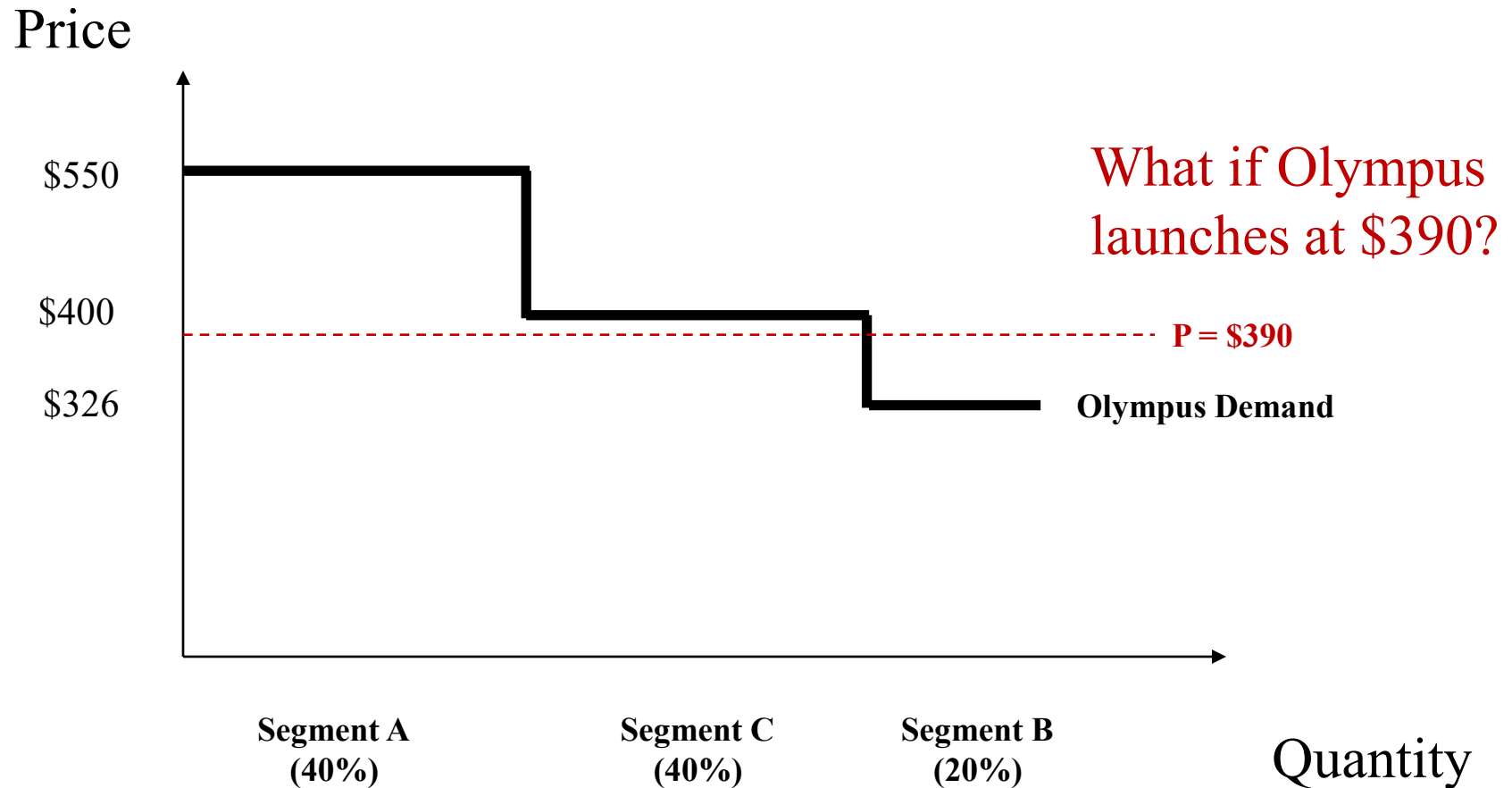
17



Olympus Demand

EVC demand when it launches against incumbents

17



Assessing Canon's Competitive Reaction

Strategy 1: Olympus at \$390

18

- First, determine Canon's new reference product in each segment
- Can just use the demand curves
- OR ...
 - ▣ If Canon not offered, will segment pick Sony or Olympus?
 - Olympus-Sony Price: $\$390 - \$250 = \$140$
 - If the Diff. Value $> \$140$ then buy Olympus
 - ▣ Diff. Value of Olympus vs. Sony
 - Seg. A: $+\$320 > \140 so buy Olympus
 - Seg. B: $+\$76 < \140 so buy Sony
 - Seg. C: $+\$187 > \140 so buy Olympus
- This gives us the Reference Brand for each segment.

Assessing Canon's Competitive Reaction

Strategy 1: Olympus at \$390

18

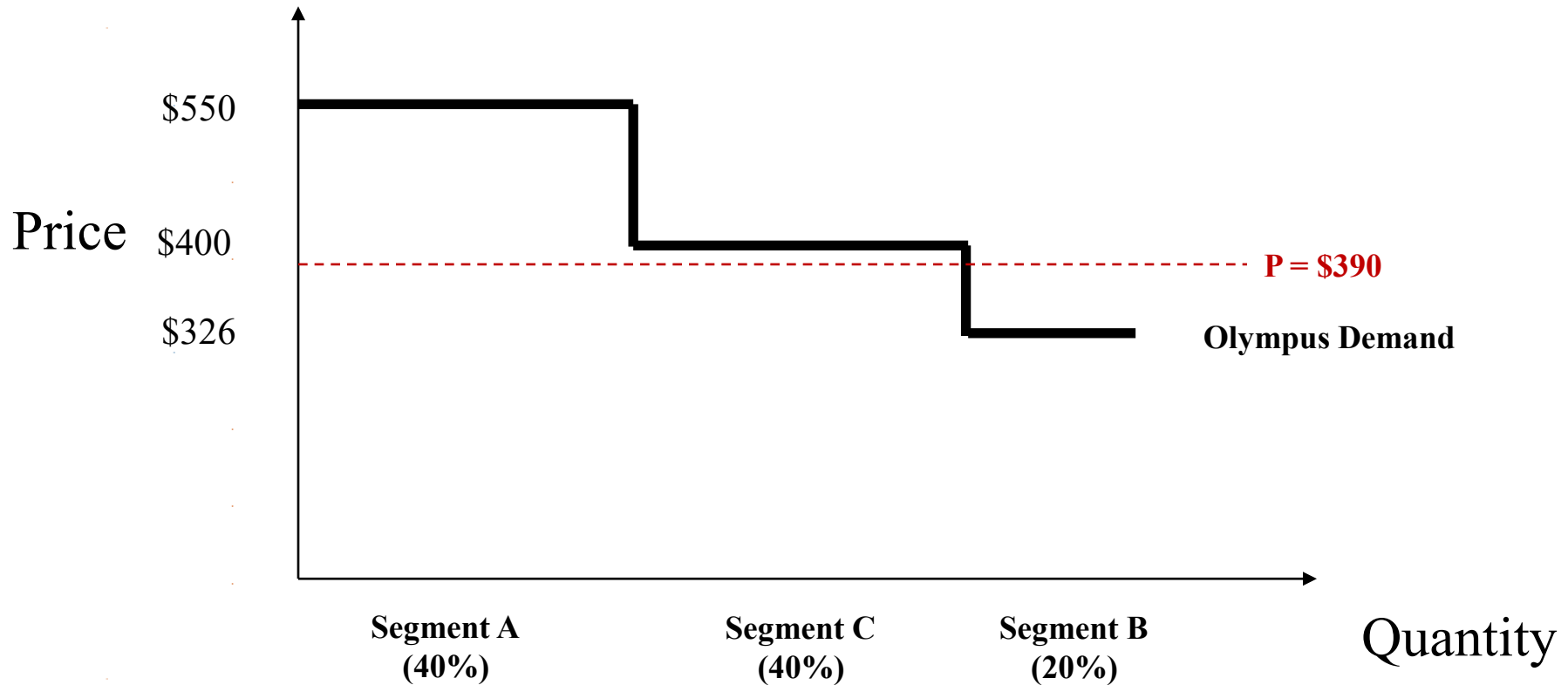
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Assessing Canon's Competitive Reaction

Strategy 1: Olympus at \$390

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Canon Demand

Sony Base at \$250, Olympus 12.2MP DSLR at \$390

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Put yourselves in shoes of Canon selling its base model

	Segment A	Segment B	Segment C
Ref Brand	Olympus	Sony	Olympus
Ref Value	\$ 390	\$ 250	\$390

Diff Value			
Meg	\$(120)	-	\$ (60)
Brand	\$(100)	\$ (40)	\$ (33)
DSLR	\$ (50)	-	\$ (27)

EVC	\$ 120	\$ 210	\$ 270
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EVC of every segment is less than \$280.
Canon has 0% share.

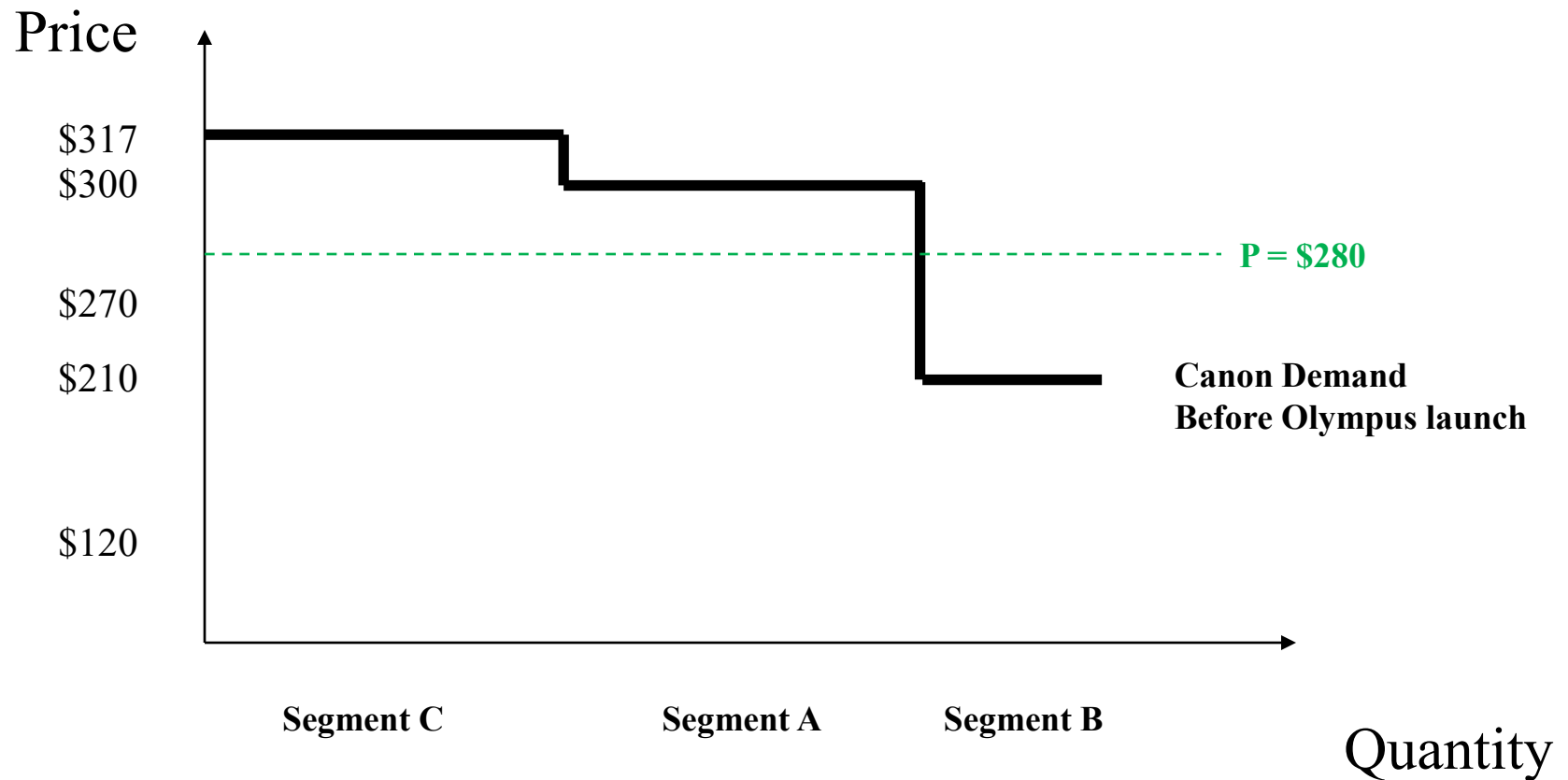
Segment B; not affected by Olympus launch, because it is higher than their EVC. Reanalyse Conjoint, and it shows Canon lost all market share to Olympus, but Sony retains 20%

This new EVC for Canon will represent the new demand curve for Canon when Olympus enters the market.

Canon Demand

EVC demand before & after Olympus (\$390)

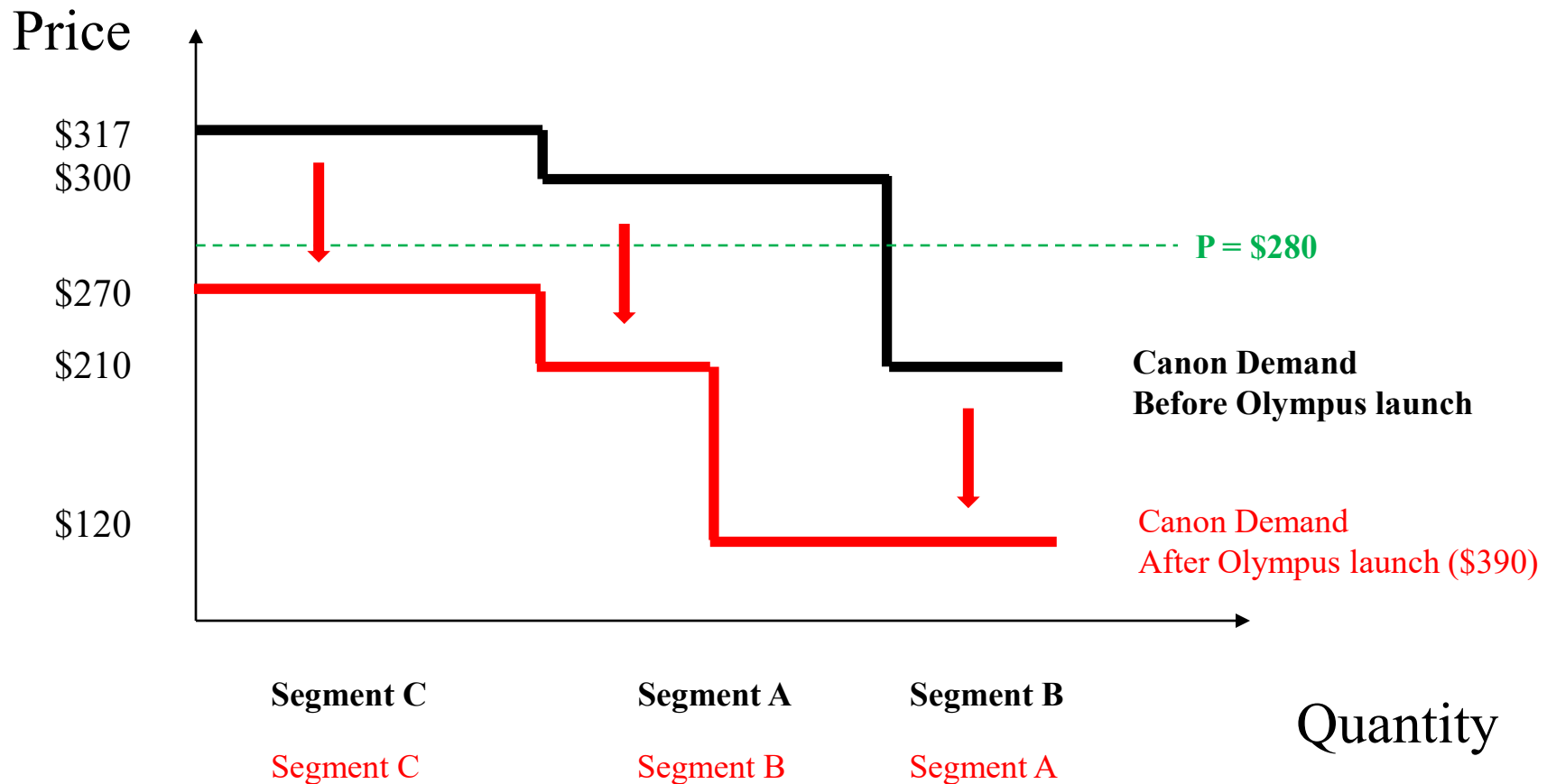
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Canon Demand

EVC demand before & after Olympus (\$390)

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Possible Price Reactions by Canon

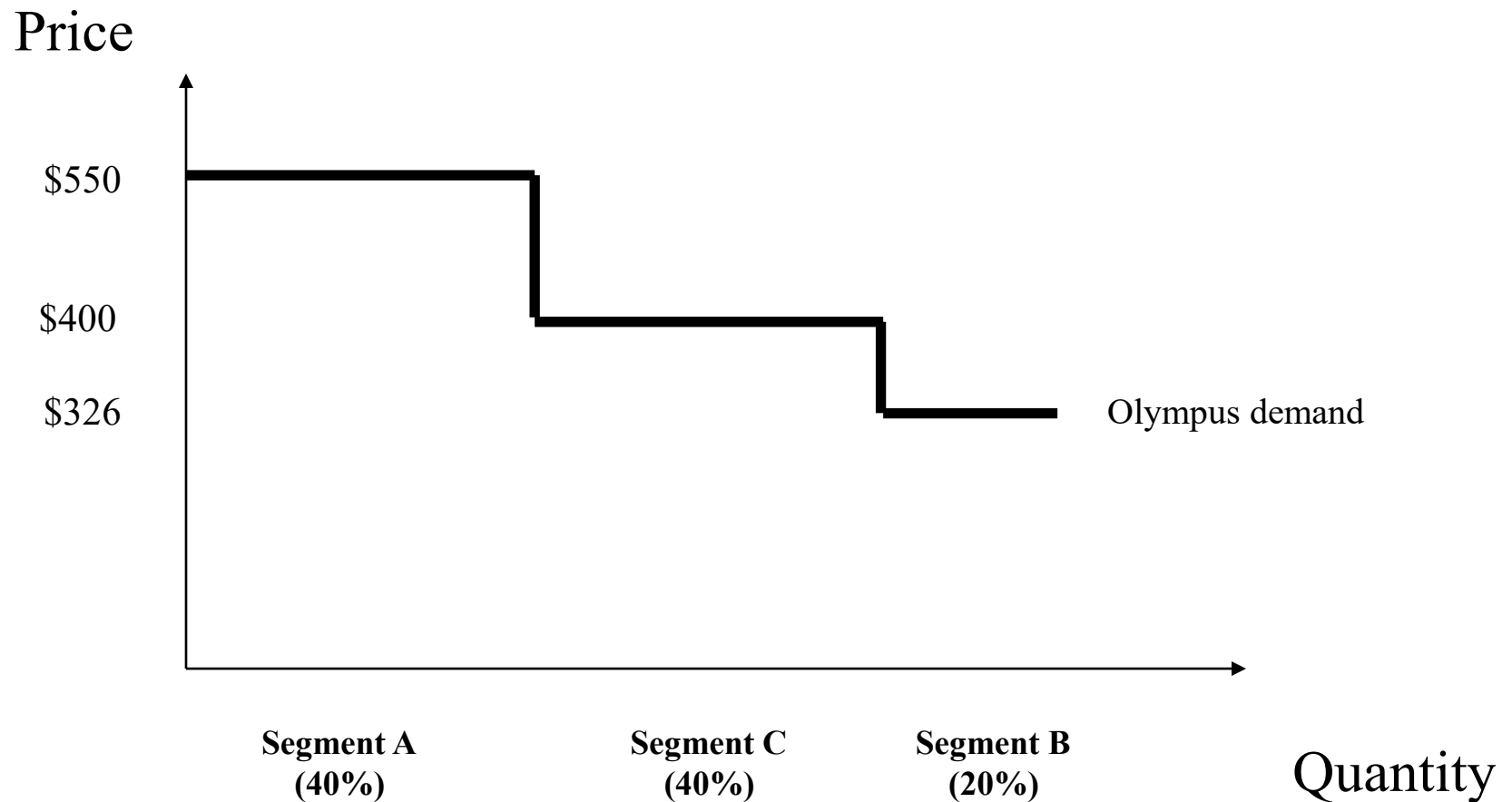
21

- \$269: attract segment C
- \$209: attract segments B and C
- \$119: attract segments A, B and C
- At a minimum, we expect to trigger a competitive reaction from Canon
- Price will fall to \$269 or lower

Olympus Demand

What if Olympus Targets Segment A?

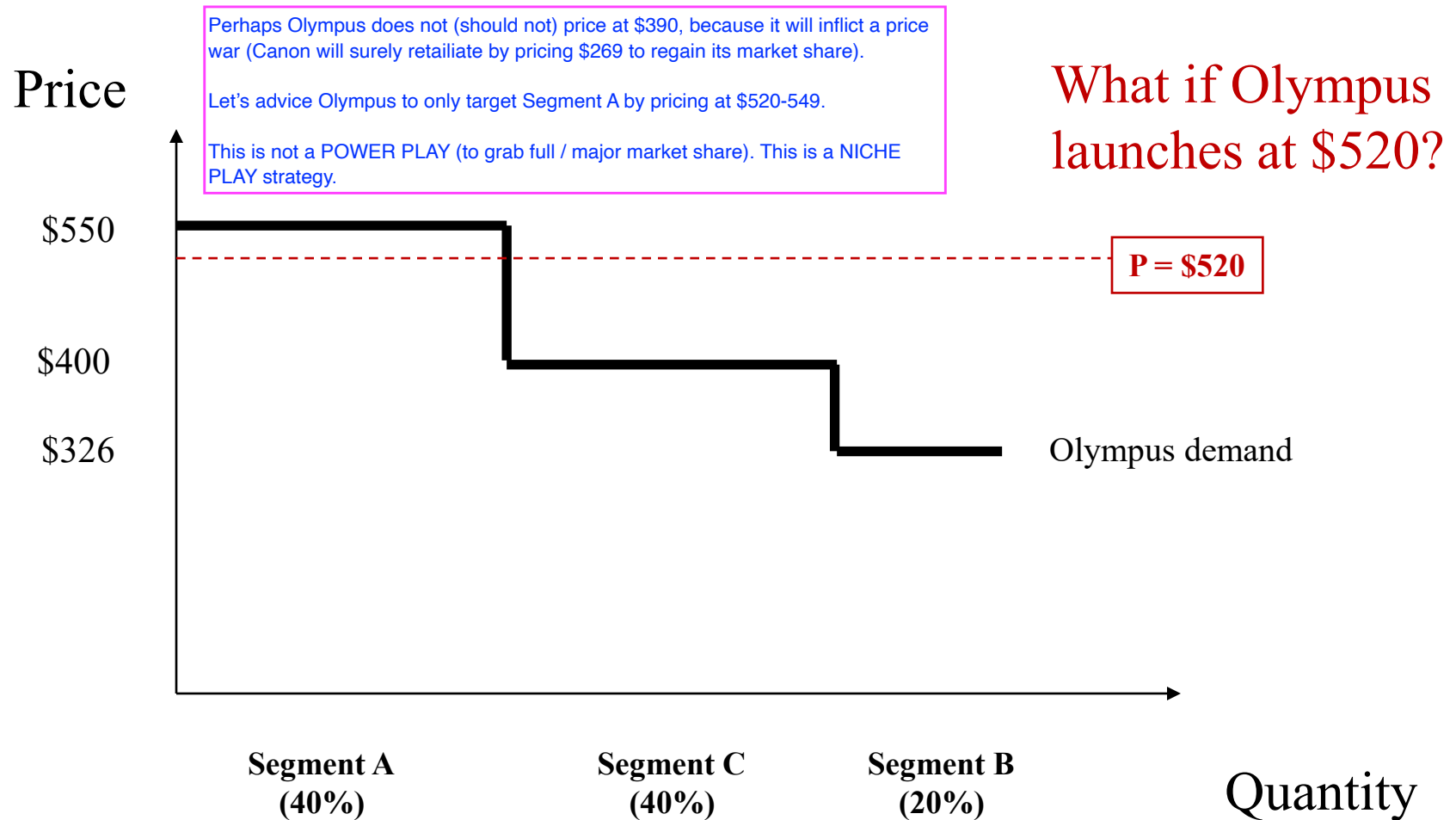
22



Olympus Demand

What if Olympus Targets Segment A?

22



Assessing Canon's Competitive Reaction

Strategy 2: Olympus at \$520

23

- First, determine Canon's new reference product in each segment
- Can just use the demand curves
- OR ...
 - ▣ If Canon not offered, will the segment pick Sony or Olympus?
 - Olympus-Sony Price: $\$520 - \$250 = \$270$
 - If the Diff. Value $> \$270$ then buy Olympus
 - ▣ Diff. Value of Olympus vs. Sony
 - Seg. A: $+\$320 > \270 so buy Olympus
 - Seg. B: $+\$76 < \270 so buy Sony
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Assessing Canon's Competitive Reaction

Strategy 2: Olympus at \$520

23

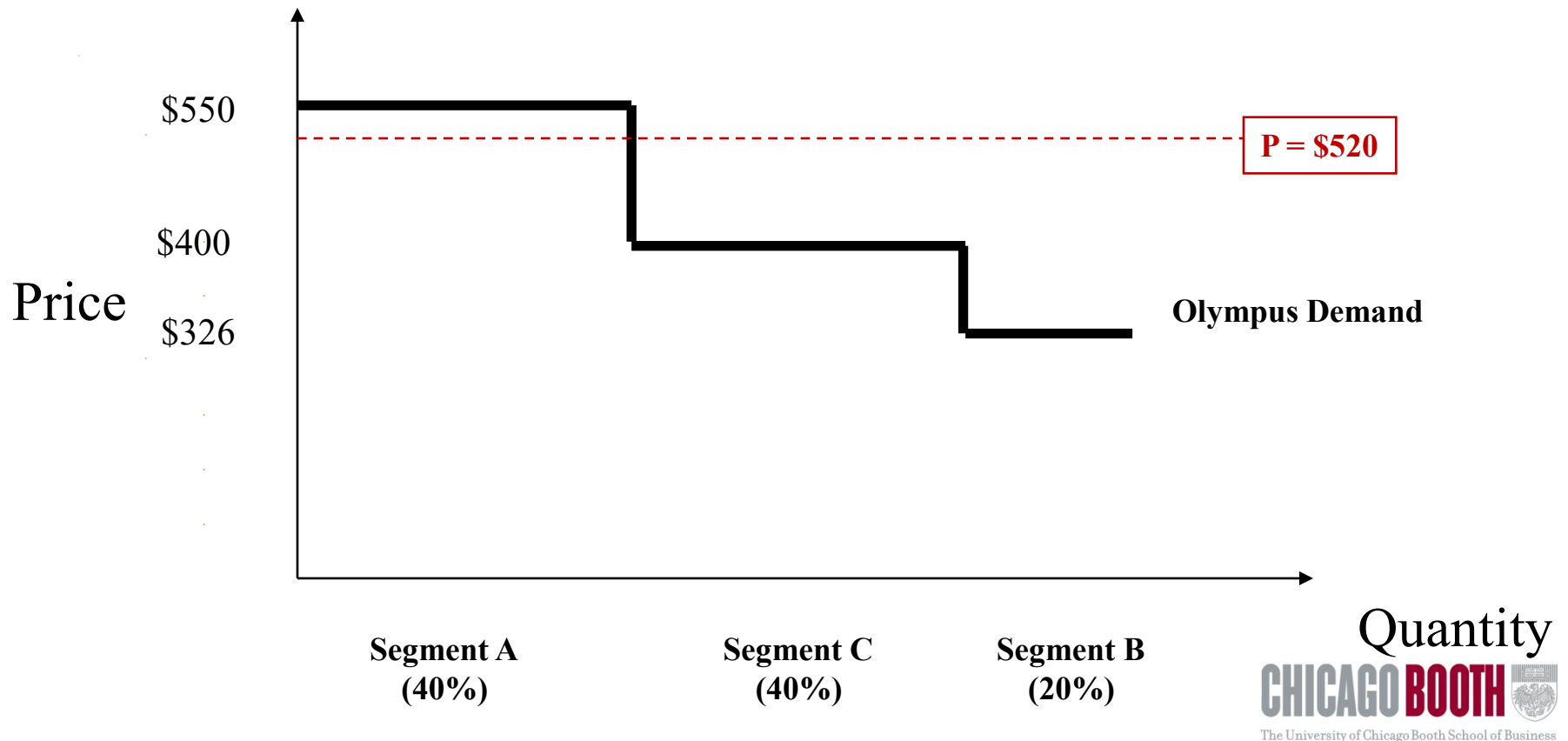
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Assessing Canon's Competitive Reaction

Strategy 2: Olympus at \$520

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Canon Base Demand

Olympus 12.2MP DSLR at \$520, Sony Base at \$250

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Put yourselves in shoes of Canon selling its base model

	Segment A	Segment B	Segment C
Ref Brand	Olympus	Sony	Sony
Ref Value	\$ 520	\$ 250	\$ 250
Diff Value			
Meg	\$(120)	-	-
Brand	\$(100)	\$ (40)	\$ 67
DSLR	\$ (50)	-	-
EVC	\$ 250	\$ 210	\$ 317

If Canon keeps price at \$280

Seg. A: buy Olympus ($\$250 < \280)

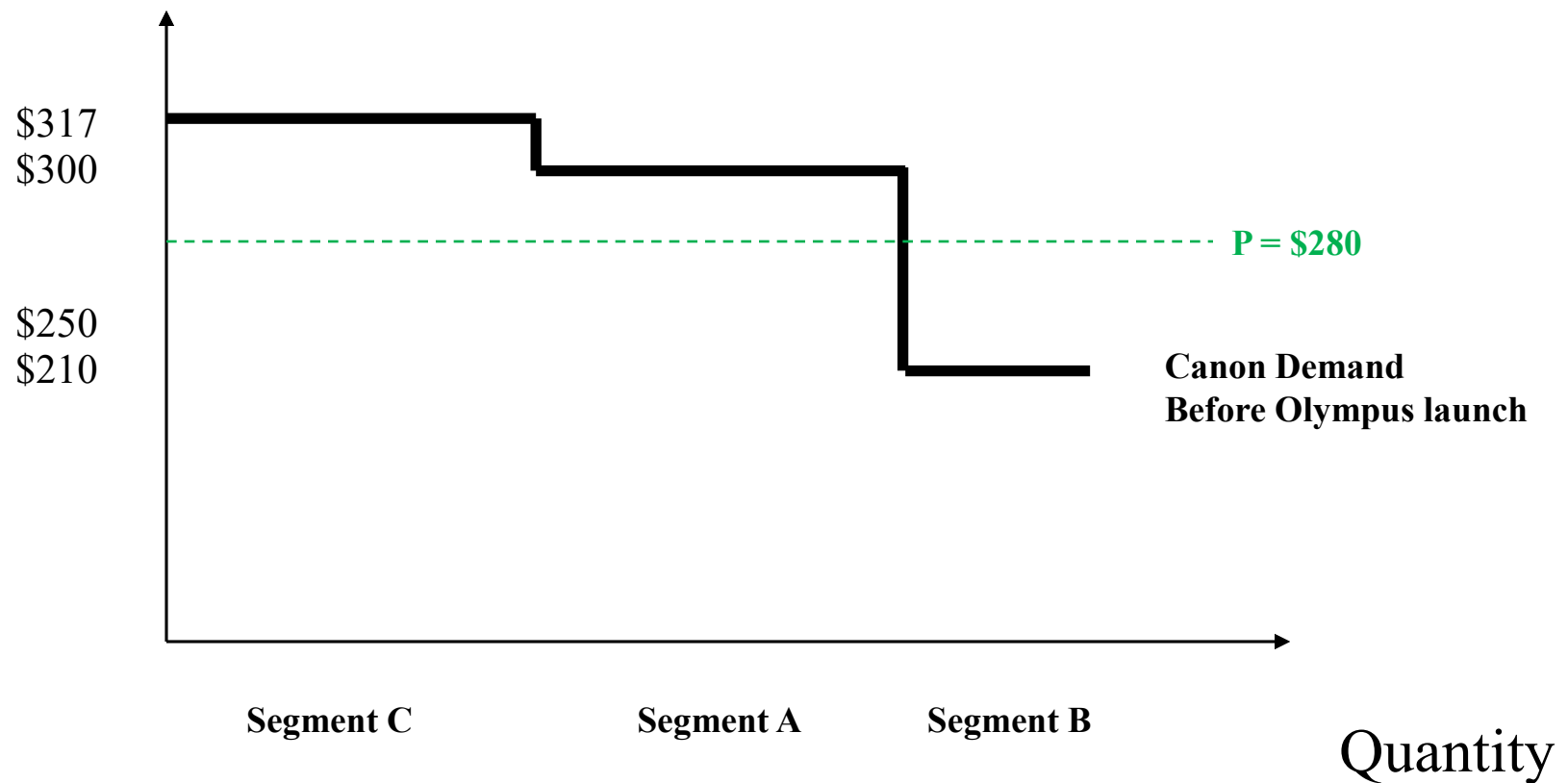
Seg. B: buy Sony ($\$210 < \280)

Seg. C: buy Canon ($\$317 > \280)

Canon Demand

EVC demand before and after Olympus (\$520)

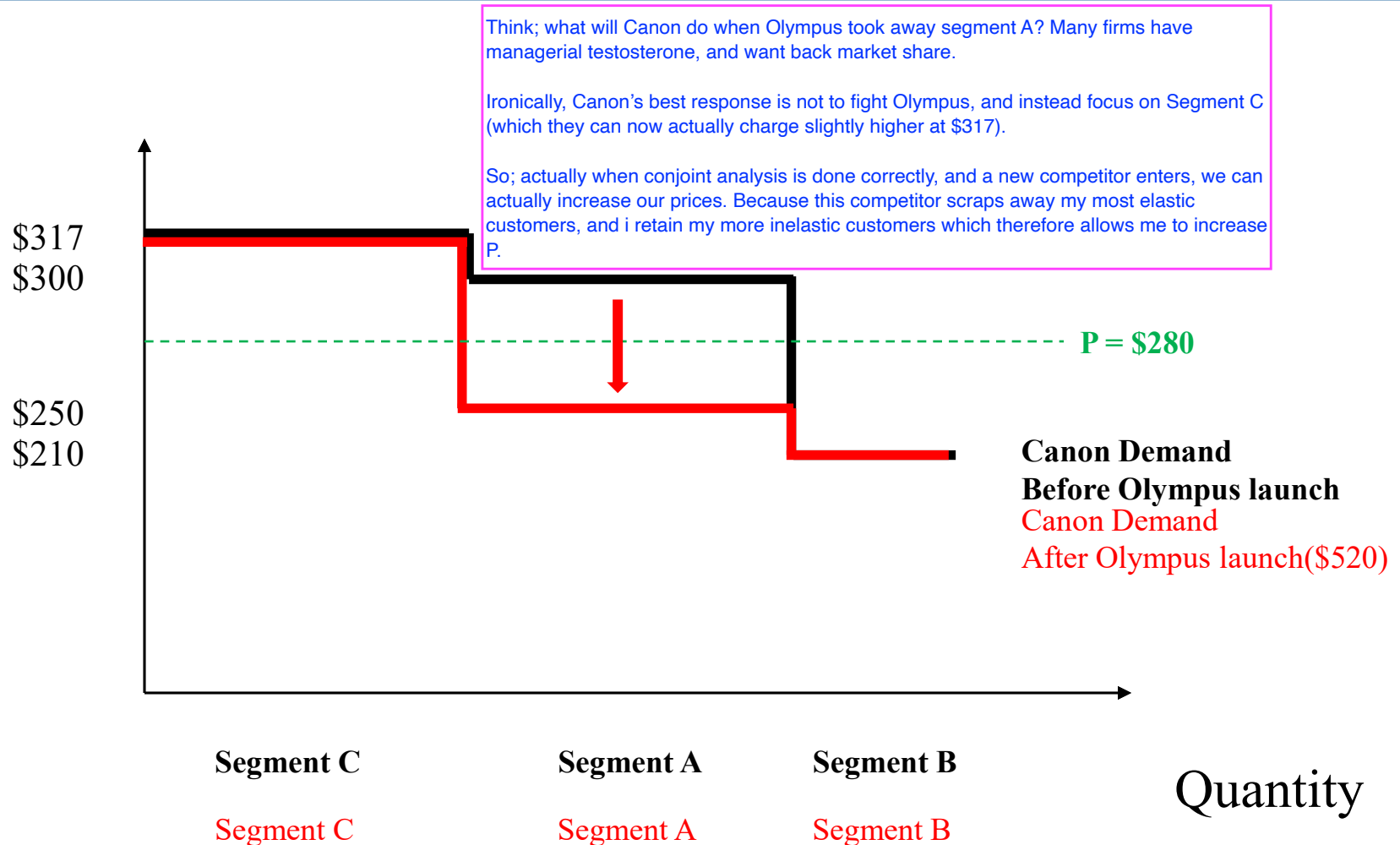
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Canon Demand

EVC demand before and after Olympus (\$520)

25



Canon Strategy

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- How large a price cut is needed to regain lost customers (i.e. segment A)?
 - ▣ \$31 price cut attracts segment A from Olympus.
 - ▣ However, price cut may trigger reaction from Olympus
- Raising Prices
 - ▣ \$37 of surplus on the table for Segment C.
 - ▣ Which firm is likely to react if Canon raises price from \$280 to \$310?

Can competition increase prices?

Examples from academic research

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- Chen and Riordan (2008): “The analysis provides precise conditions for price-increasing competition and reveals that it is unexceptional from a theoretical perspective.”
- Ward et al (2002): “The conventional industry wisdom is that namebrand firms defended their brands against the private-label invasion by lowering their prices, engaging in additional promotional activities, and increasingly differentiating their products. Our empirical evidence is inconsistent with these beliefs. Increases in the share of private-label goods are correlated with a rise in the price of namebrand goods.”
- Grabowski and Vernon (1992): “We performed an analysis of eighteen drugs first experiencing generic competition after 1983. These products have been subject to a wave of generic competition leading to a significant loss in market share to generics that sell at much lower unit prices...The pioneering firms did not attempt to deter entry through their pricing strategies. Rather, in most cases, the firms continued to increase their prices at the same rate as prior to entry.”

Can competition increase prices?

Example: Public Pharmacies in Chile

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- Public pharmacies entered 147 of 345 counties in Chile between 2015 and 2018
 - ▣ Lower costs (due to public buying platform)
 - Prices = 34% of prices charged by private pharmacies!
 - ▣ But Lower Quality: less variety and limited hours open

Private Pharmacy



Public Pharmacy



Can competition increase prices?

Example: Public Pharmacies in Chile

29

- Public pharmacies diverted 4% of market share away from private pharmacies
- But...

Private pharmacies responded by increasing prices 1.1%

- Most price-elastic consumers switched to public, leaving an overall less elastic residual demand for private!

Competition, Game Theory & Equilibrium Prices



The Strategic Pricing Game

Beauregard Textiles

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- Elasticity
- Profits: Revenues, Costs
- Payoffs



Payoff Matrix

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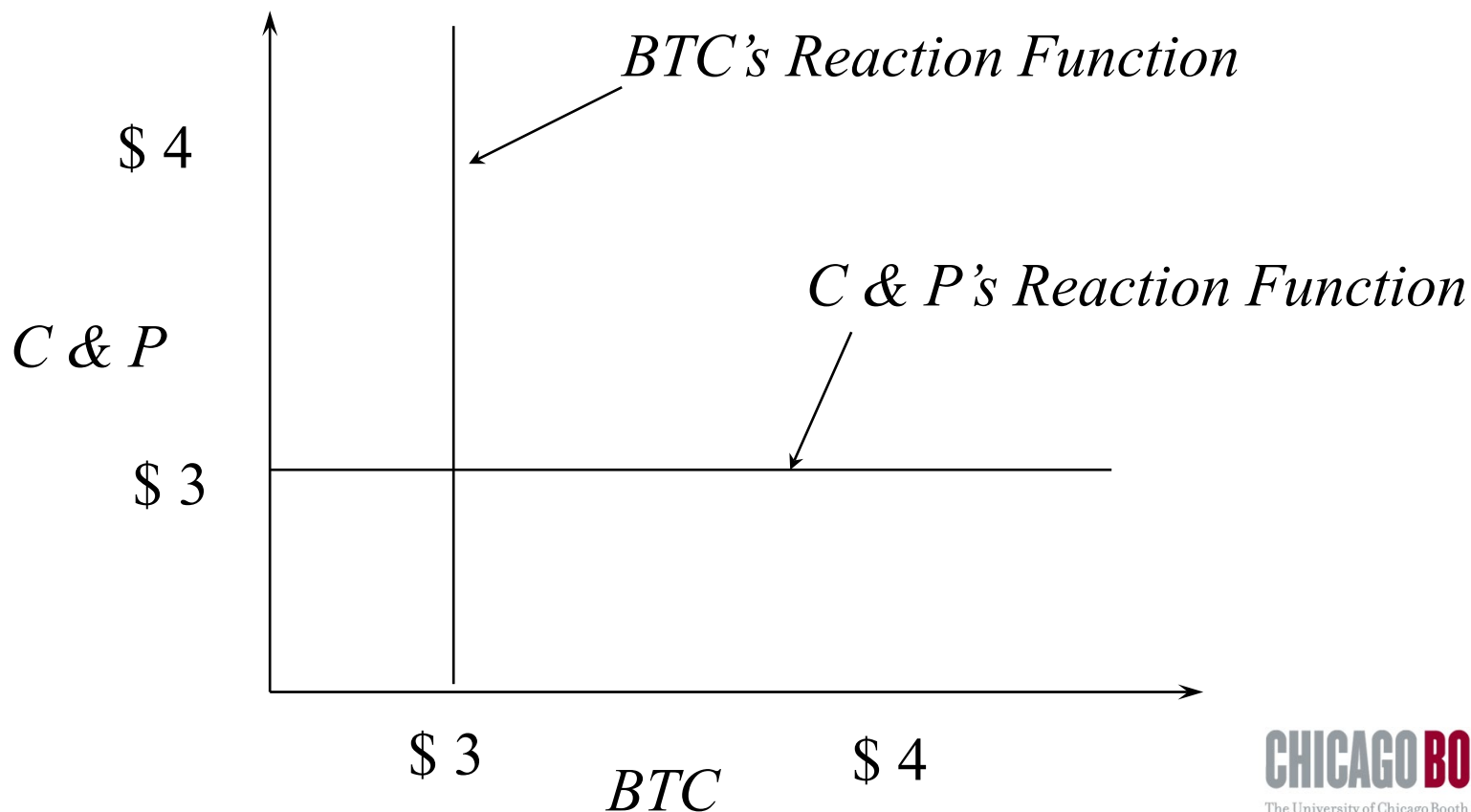
		C&P	
		\$ 3	\$ 4
BTC	\$ 3	C&P: \$ 166,800 BTC: \$ 212,550	C&P: \$ 62,500 BTC: \$ 332,000
	\$ 4	C&P: \$ 258,300 BTC: \$ 198,000	C&P: \$ 211,200 BTC: \$ 266,800

See appendix for details.

Reaction Functions

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- For every possible price by BTC, what is C&P's best price?
- For every possible price by C&P, what is BTC's best price?



Equilibrium

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- For both BTC and C&P, choosing \$ 3 is the optimal strategy, irrespective of the competitor's price
 - ▣ \$ 3 is a *dominant strategy* for both BTC and C&P
- Nash equilibrium: (BTC: \$3, C&P: \$3) -- neither BTC nor C&P will deviate from \$3 pricing strategy

Raising Industry Prices

35

- In the Nash equilibrium, pay-offs of BTC and C&P are:
 - BTC: \$ 212,550 C&P: \$ 166,800
 - Total Industry Profit: \$ 379,550

- If both BTC and C&P could commit to price at \$4, pay-offs of BTC and C&P are:
 - BTC: \$ 266,800 C&P: \$ 211,200
 - Total Industry Profit: \$ 478,000

Raising Industry Prices

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➔ Prisoner's Dilemma



See Appendix for Black Friday Sales as example of a prisoner's dilemma

Ethical concerns with Price Wars as a prisoner's dilemma: Cartels

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- Price Fixing is Illegal (see [FTC](#))
 - ▣ You will get fined and/or go to jail
 - ▣ But some firms will nevertheless try to coordinate illegally

“Pilgrim's Pride pleads guilty to chicken price-fixing charge, to pay \$107.9 mln fine”

Jonathan Stempel, Reuters, 2-23-2021

“U.S. grand jury indicts former Pilgrim's Pride executives for chicken price fixing”

Reuters, 7-30-2021

“A federal judge partially dismissed a shareholder's suit against Pilgrim's Pride on Tuesday, removing the company's current CEO from litigation but advancing claims against the company and its former CEO.”

WSJ, 12-23-2023

Ethical concerns with Price Wars as a prisoner's dilemma: Cartels

36

- Price Fixing is Illegal (see [FTC](#))
 - ▣ You will get fined and/or go to jail
 - ▣ But some firms will nevertheless try to coordinate illegally

“Pilgrim's Pride pleads guilty to chicken price-fixing charge, to pay \$107.9 mln fine”

Jonathan Stempel, *Reuters*, 2-23-2021

“U.S. grand jury indicts former Pilgrim's Pride executives for chicken price fixing”

Reuters, 7-30-2021

“A federal judge partially dismissed a shareholder's suit against Pilgrim's Pride on Tuesday, removing the company's current CEO from litigation but advancing claims against the company and its former CEO.”

WSJ, 12-23-2023

- But even if it were legal...

Threat of Punishing in Beauregard case

37

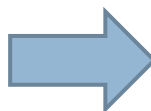
	BTC	C&P
Cooperate	266,800	211,200
Defect	332,000	258,300
Punish	212,550	166,800
Gain from Defect	65,200	47,100
Loss from Punish	-54,250	-44,400

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Even if BTC & C&P coordinated, gains from defection exceed incentives to cooperate.

 Coordination difficult to sustain.

Cartel Instability

Example: The French Laundry Detergent Market

38

“Dirty Secrets in Soap Prices”

Max Colchester and Christina Passariello

Wall Street Journal, 12-9-2011

“For nearly a decade, executives from consumer goods giants including Proctor & Gamble Co. secretly met in discreet restaurants around Paris, allegedly to fix the price of laundry detergent in France.”



Cartel Instability

Example: The French Laundry Detergent Market

39

- In 1996, four brand managers met to ensure they all charged pre-agreed prices and would alert one-another to deals.

Cartel Codenames

French authorities detailed an alleged price-fixing scheme by major companies complete with code names and secret meetings.

CODENAME:

'Pierre'

'Laurence'

'Hugues'

'Christian'

COMPANY:

P&G


Unilever




COLGATE-PALMOLIVE

Caught in flagrante delicto

40

- “Team” caught colluding at Paris Brasserie “Le Royal Villier” during a 2006 raid



Cartel Instability

Example: The French Laundry Detergent Market

41

- Disagreements broke down the arrangement in 2004:
 - ▣ Unilever cut price 10%
 - ▣ P&G responded with 25% price cut
 - ▣ Henkel finally responded with 40% price cut.

Cartel Instability

Example: The French Laundry Detergent Market

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- **Epilogue:** France fines 13 consumer goods firms €951m for price-fixing (Betsy Reed, [The Guardian](#), Dec. 18, 2014)



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- **Epilogue:** France fines 13 consumer goods firms €951m for price-fixing (Betsy Reed, [The Guardian](#), Dec. 18, 2014)
 - ▣ Henkel granted leniency for testimony



Back to algorithmic pricing ...

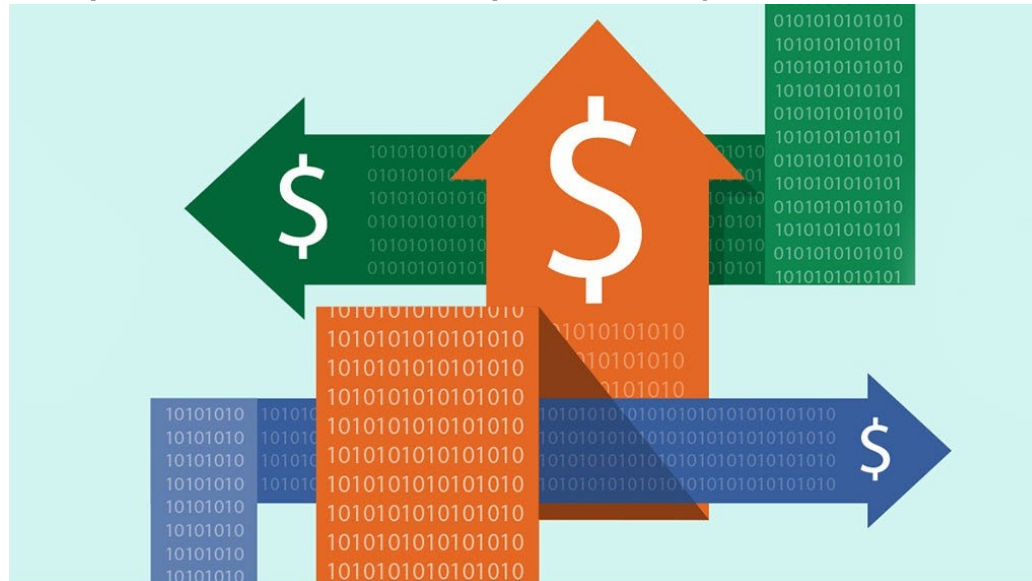
42

- Use of **algorithms** implicitly commits to irrational pricing “out of equilibrium” (i.e., instant punishments)



Collusive pricing outcomes without any actual collusion

- But if a manager over-rides algorithm because they know something algorithm cannot predict, prices become competitive again



Take Away

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- Your price decisions may change competitor's pricing incentives!
- Using EVC to assess reaction
 - ▣ **Think like your competitor**
- Need equilibrium definition to make price predictions with strategic interactions (see competitive strategy course)
- Algorithmic pricing may soften price competition

dont leave money on the table. And your pricing may change competitor pricing incentives.

Competition can sometimes lead to higher price

See optional Appendix for how to conduct a numerical market simulation for two competing firms when demand has been estimated and the Nash equilibrium can be **computed**.

Details on payoff matrix for Beauregard Textiles

Own Elasticity

45

Beauregard Sales	Beauregard Price	C & P Price
125,302	\$ 3.00	\$ 3.00
74,860	\$ 4.00	\$ 3.00
	% Q	-40%
	% P	33%
Own Price Elasticity		(1.21)

Notice: C&P Price Constant

Regression: Own Elasticity = -1.75

Cross Elasticity

46

Beauregard Sales	Beauregard Price	C & P Price
125,426	\$ 3.00	\$ 3.00
198,863	\$ 3.00	\$ 4.00
	% Q	59%
	% P	33%
Cross Price Elasticity		1.76

Notice: Beauregard Price Constant

Regression: Cross Elasticity = +1.59

Profits

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BTC

Price	\$4.00		
Direct Labor	0.80		
Material	0.40		
Mat'l Spoilage	0.04		
Dir Dept Exp	0.12		
Total Relevant Costs		\$1.36	
Total Contribution at 75,000 yds			\$198,000

C&P

Price	\$3.00		
Direct Labor	0.740		
Material	0.400		
Mat'l Spoilage	0.038		
Dir Dept Exp	0.100		
Total Relevant Costs		\$1.278	
Total Contribution at 150,000 yds			\$258,300

Payoffs

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- The cells in the payoff matrix are computed by using the cost data and predicted sales
- Use elasticities to predict sales

Black Friday as a Prisoner's Dilemma

Prisoners' Dilemmas in Pricing

FIRST to offer discounts to holiday shoppers

50

- Retailers Face “Prisoners' Dilemma” by Opening on Thanksgiving
- Black Friday Deals...What's all the fuss?

Jean-Pierre Dubé, Sigmund E. Edelstone Professor of Marketing at the University of Chicago Booth School of Business joins Hannah and Dane to talk about how Black Friday deals are slowly losing their allure.

WGN, 11-27-2015

- Evolution of the “opening early on Thanksgiving” game
 - ▣ 2011: Toys ‘R Us opened at 9pm
 - ▣ 2012: Target and Lord & Taylor
 - ▣ 2013: Macy’s announced would open at 8pm
 - Target opened at 8pm
 - Walmart and Best Buy opened at 6pm
 - Toys ‘R Us opened at 5pm

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- Net benefit? ... Costs?

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Using R to find Nash equilibrium

Case Study

A hypothetical new competitor to Ziprecruiter

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- You have been hired by Ziprecruiter to analyze how its pricing power has changed after entry by a new competitor.
- The new competitor has less visibility but also has a reasonably competitive platform.
- Using web-browser data, you have already conducted an empirical analysis to determine demand for Ziprecruiter and demand for its new competitor.
- Now, you want to analyze the Nash equilibrium in prices using numerical tools in R

Empirical Analysis of Demand

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- As in module 2 appendix, use logistic model of demand. Now demand is trinomial (not binomial) since prospective customers can choose (1) Zip, (2) competitor, or (3) not purchase.

- Logit shares:

$$S_Z(p_Z, p_C) = \frac{\exp(-.6 - .004p_Z)}{1 + \exp(-.6 - .004p_Z) + \exp(-.3 - .004p_C)}$$

$$S_C(p_Z, p_C) = \frac{\exp(-2 - .004p_C)}{1 + \exp(-.6 - .004p_Z) + \exp(-2 - .004p_C)}$$

where p_Z and p_C are the prices for Ziprecruiter and competitor

Nash Equilibrium

54

- The definition of Nash equilibrium in this case implies that both firms are simultaneously setting a price that *maximizes its profits*.
- In other words, each firm is setting a price that satisfies its inverse-elasticity rule:

$$p_Z = \frac{\text{elas}_Z(p_Z, p_C)}{1 + \text{elas}_1(p_Z, p_C)} mc_Z$$

$$p_C = \frac{\text{elas}_C(p_Z, p_C)}{1 + \text{elas}_C(p_Z, p_C)} mc_C$$

where unlike our log-log demand models, the own-price elasticity is now a function (as opposed to a constant number). Also, mc_Z and mc_C are Zip's and competitor's marginal costs.

Nash Equilibrium

Information Goods

55

- Zip and competitor face zero marginal costs because their service is digitized: $mc_Z = mc_C = 0$
- From module 2 appendix, we know we can simplify the pricing rules to:

$$\text{elas}_Z(p_Z, p_C) = \text{elas}_C(p_Z, p_C) = -1$$

i.e., a Nash equilibrium for these two *information-goods services* will be a pair of prices where both firms are pricing on the inelastic region of demand.

Own Price Elasticity

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- To compute an own-price elasticity, use the formula:

$$\begin{aligned}\text{elas}_Z(p_Z, p_C) &= \frac{\partial S_Z(p_Z, p_C)}{\partial p_Z} \frac{p_Z}{S_Z(p_Z, p_C)} \\ &= -0.004(1 - S_1(p_1, p_2))p_1\end{aligned}$$

See if you can derive this result using calculus and a pen & paper.

Similarly, the elasticity for competitor is:

$$\text{elas}_C(p_Z, p_C) = -0.004(1 - S_C(p_Z, p_C))p_1$$

Computing the Nash Equilibrium

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To solve for the Nash equilibrium, we want to find prices p_Z^* and p_C^* such that

$$\text{elas}_Z(p_Z^*, p_C^*) = \text{elas}_C(p_Z^*, p_C^*) = -1$$

Don't get intimidated ... this isn't as hard as it may seem...

R code to compute the Nash equilibrium

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□ Function to measure difference in each elasticity and “-1”:

```
➤ ElasRuleComp = function(x) {  
➤   denom = 1+exp(-.6-.004*x[1])+exp(-2-.004*x[2])  
➤   Sz = exp(-.6-.004*x[1])/denom  
➤   Sc = exp(-2-.004*x[1])/denom  
➤   DElasZ = -0.004 * (1-Sz)*x[1]+1  
➤   DElasC = -0.004 * (1-Sc)*x[2]+1  
➤ }
```

□ Load the solver function in R “nleqslv”

```
➤ library(nleqslv)
```

□ Now use R solver function “nleqslv” to solve for (pz*,pc*):

```
➤ out=nleqslv(x=c(0,0),fn=ElasRuleComp)
```

□ Extract optimal price:

```
➤ p.star.equilib = out$x
```

Explore

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- As you raise and lower the intercept for the competitor, you accordingly raise and lower how viable a competitor it is.
- Why not check to see if your pricing results change much as you “tweak” the parameters. As you “tweak” them, consider what that means substantively (i.e., you are changing the implied customer preferences)